

Heaven's Light is Our Guide



Rajshahi University of Engineering & Technology

Department of Mechatronics Engineering

Course No: MTE 3200

Course Title: Mechatronics Case Study

A Project Report on:

“Unmanned Surface Vehicle for Flood Search and Rescue”

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Declaration

We, Fahim Faisal (2008001), Fahad Bin Islam (2008002), Nishat Pervez Meghoboti (2008003), Ariful Islam Riad (2008004), and Hodayra Mobesshira Hemu (2008005), certify that we are the only authors of this report on a case study entitled "Unmanned Surface Vehicle for Flood Search and Rescue." The report is submitted to the Department of Mechatronics Engineering of Rajshahi University of Engineering & Technology (RUET) in partial fulfillment of our B.Sc. degree requirements.

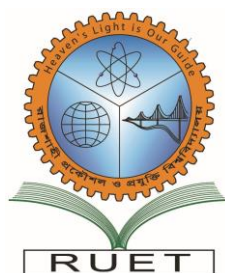
The research work presented here was undertaken under the guidance of Sumaya Ishrat Moyeen, Assistant Professor, Department of Mechatronics Engineering, RUET. All technical ideas, design strategies, data, and results that have evolved in the course of the project are the authors' sole contributions, unless otherwise mentioned.

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CERTIFICATE

This is to certify that the project report titled “Unmanned Surface Vehicle for Flood Search and Rescue” by Fahim Faisal (2008001), Fahad Bin Islam (2008002), Nishat Pervez Meghoboti (2008003), Ariful Islam Riad (2008004), and Homayra Mobesshira Hemu, (2008005) has been carried out under my supervision in partial fulfilment of the requirements for the degree of Bachelor of Science in Mechatronics Engineering at Rajshahi University of Engineering & Technology (RUET).

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Acknowledgement

Unmanned Surface Vehicle Flood Search and Rescue. This report is submitted to the Department of Mechatronics Engineering at Rajshahi University of Engineering & Technology (RUET) in compliance with our 4 year B.Sc. degree requirements.

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We acknowledge and respect the value of academic integrity and realize that any acts of plagiarism or any other misrepresentation of work could lead to serious academic consequences. In the event of any doubts on the validity of this report we are willing to offer additional explanation or evidence.

Abstract

This is an integrated case study of the design and development of an Unmanned Surface Vehicle (USV) for search and rescue missions in flood disasters. This was an engineering project that was carried out with the purpose of addressing real problems of disaster rescue in flooded districts. The central goal was to design an autonomous boat that can navigate through flooded areas, capture real-time position data, and assist in locating victims in distress without compromising human lives.

With an eight-week development cycle, the project onboarded significant mechatronics and embedded system components including the ESP32 microcontroller, GPS module, gyroscope sensor, motor driver, DC motors and servo motor. The team set out to overcome formidable engineering challenges such as real-time navigation, remote monitoring, sensor fusion, waterproofing, and system stability.

Through iterative prototyping, simulation, coding, and testing, the project offered enriching hands-on education in multidisciplinary engineering design and problem-solving. The report combines observations, technical takeaways, and outcomes of the development phases, highlighting the effectiveness of low-cost autonomous systems in disaster scenarios. While the project was fraught with issues including environmental simulation limitation and hardware calibration issues, it enhanced the participants' skills and knowledge in embedded systems, robotics, and green engineering practices tremendously. The study also contributes to the international conversation on the use of smart, unmanned platforms to support emergency response and community resilience.

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List of Abbreviations

USV	-----	Unmanned Surface Vehicle
ASV	-----	Autonomous Surface Vehicle
UAV	-----	Unmanned Aerial Vehicle
UGV	-----	Unmanned Ground Vehicle
SAR	-----	Search and Rescue
FSM	-----	Finite-State Machine
EKF	-----	Extended Kalman Filter
IMU	-----	Inertial Measurement Unit
ROS 2	-----	Robot Operating System 2
GA	-----	Genetic Algorithm
A*	-----	A-star (pathfinding algorithm)
PRM	-----	Probabilistic Roadmap (path planning algorithm)
IP67	-----	Ingress Protection Rating 67 (dust-tight and water-resistant up to 1 meter)
ESP8266	-----	Espressif Systems Programmable Wi-Fi Microcontroller 8266
ESP32	-----	Espressif Systems Programmable Wi-Fi/Bluetooth Microcontroller 32-bit
NEO-6M	-----	A GPS Module by u-blox (model NEO-6M)
SG90	-----	Standard Small Servo Motor (SG90 model)
VTOL	-----	Vertical Take-Off and Landing
LoRa	-----	Long Range (low-power wide-area network communication protocol)

Chapter 1

Introduction

1.1 Introduction

With the growing frequency of water calamities and sea distress, the need for an autonomous rescue system has become highly critical. Traditional rescue efforts involve human risk, high operating cost, and latency in the event of environmental challenges. Autonomous vehicles, especially rescue boats, present a feasible solution to this scenario. This project offers an "Autonomous Rescue Boat" that is capable of traveling autonomously to a given GPS location with the assistance of an ESP8266 microcontroller and a NEO-6M GPS module. The boat is suggested to assist in emergency operations like river floods, fishing accidents, or coastal patrol without incessant human interference.

1.2 Background Studies

There has been a sudden surge in the use of autonomous technology in recent years across various industries, including transportation, agriculture, and the military. Autonomous surface vehicles (ASVs) are employed in naval regions for surveying, patrol, and delivery.[1][2]

ESP8266, a Wi-Fi-enabled microcontroller, has proven to be ideally placed for IoT projects due to its low cost and adaptability. With NEO-6M GPS module, real-time location tracking is possible. The system can be programmed to go to a location by entering latitude and longitude coordinates with very little drift [3][4][5][6].

Other notable projects:

- RF- and GSM-powered rescue boats[17]
- GSM- and RF-powered search and rescue drones
- GPS-led robots powered by Arduino[12][13]

Our project builds on these foundations by integrating live GPS direction with wireless command and autonomous decision-making into a simplified prototype in the complex operational environment of flooded areas like the ones recently got flooded in Feni and Noakhali.[14][19][20]

1.3 Motivation

- Flood in Feni submerged > 70% of low-lying land, affecting ~ 20,000 rendering people homeless.
- Houses damaged Exceeding 5,000 houses were damaged and late rescue efforts left in the lurch for hours.
- The 2023 Noakhali flood impacted 150,000+ and disrupted major transportation roads.
- In some places, clock time had passed 12 hours for the traditional rescues.
- Debris-choked waters and powerful currents were an imminent threat to human rescuers.
- More than 30 of the reported injuries last year took place as part of flood relief operations.

1.4 Objectives

The primary goals of this project are:

- To create and implement an autonomous rescue boat using ESP8266 and NEO-6M GPS.
- To give real-time GPS-based navigation to a destination point (predefined coordinates).
- To show autonomous correction of path using heading direction and feedback.
- To give future prospects for adding sensors (e.g., gas leak, temperature, drowning detection).

Secondary goals:

- Design an Android-based GUI interface to input GPS coordinates.
- Give real-time location feedback from the boat to the user.
- Provide safe, stable, and power-effective operations on the water.

1.5 Work Plan

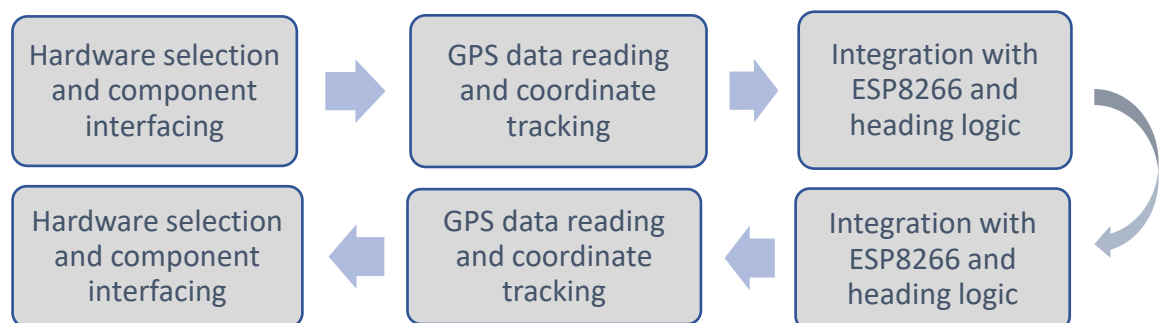


Figure 1.1 Work Plan flow chart

1.6 Organization of the Report

This project report has been organized in eight chapters, one for each significant detail of the Autonomous Rescue Boat project. The chapter-wise structure is as follows:

Chapter 1: Introduction

Describes the background, motivation, objectives, and preliminary planning of the project, along with a Gantt chart illustrating the timeline.

Chapter 2: Literature Review & Methodology

Describes existing literature on autonomous boats and GPS navigation systems, and also discusses gaps in the research, methodology and system design.

Chapter 3: Design & Implementation

Discusses the technical design process, hardware and software development, and performance standards.

Chapter 4: Project Management & Finance

Explains the planning, equipment acquisition, time management, and finance required to execute the project effectively.

Chapter 5: Addressing Complex Engineering Problem & Activities

Highlights how the project deals with real-life complex engineering problems and activities involving multidisciplinary knowledge and critical decision-making.

Chapter 6: Result & Analysis

Provides test results and performance metrics, along with a comparison of predicted and actual behavior of the system.

Chapter 7: Socio-economic Impact on Sustainability

Discusses the social, ethical, environmental, and economic implications of the autonomous rescue boat.

Chapter 8: Conclusion

Summarizes the most important findings, projects limitations, and proposes directions of future development.

References

Lists all the books, research papers, articles, and web-based sources mentioned in the report throughout.

1.7 Summary

This chapter presented an overview of an Autonomous Rescue Boat that may go to specific GPS coordinates using an ESP8266 microcontroller and NEO-6M GPS module. We explained the purpose of the project, objectives, and significance of such a system in disaster-prone regions like Bangladesh. A work plan in detail as well as chapter-wise outline of the structure of the report was also presented. The next chapter presents related works, identifies the limitations of current systems, and discusses the approach that has been used to design a reliable, efficient, and cost-effective rescue mechanism.

Chapter 2

Literature Review & Methodology

2.1 Introduction

In recent years, for the increasing frequency and intensity of flood due to climate change have emphasized the need for more effective and timely search and rescue (SAR) operations. Traditional approaches applied in flood response have often been complicated with problems of inaccessibility, longer, and risk to human rescuers. Therefore, autonomous Unmanned Surface Vehicles (USVs) is gaining attention in search and rescue efforts in flooded areas. USVs are remote-controlled or autonomous boats that are able to work in flooded areas to search for, assist or transfer victims, and to gather real-time data. This literature review discusses the ultramodern research and development of USVs for flood SAR in the aspects of design architectures, sensor integration, autonomous navigation systems, communication systems, and practical applications. The review also identifies gaps in existing systems and indicates potential areas where the performance and safety of USVs in dynamic and hazardous flood environments can be improved.

2.2 Literature Survey

Table 2.1 Literature Survey

Title	Method Used	Result	Future Work	Journal/ Conference name	Pub lica tion yea r
Obstacle Detection from Overhead Imagery using Self-Supervised Learning for Autonomous Surface Vehicles	The paper combines sonar data from Echo Park Lake and King Harbor with overhead imagery processed using Gabor filters to create obstacle maps for ASVs.	It addresses challenges like misclassification of tree shadows, leading to more accurate mapping result	Future work involves using a probabilistic framework to handle multiple labels, occlusions (e.g., from bridges/trees), and sequential data for improved classification.	2011 IEEE/RSJ International Conference on Intelligent Robots and Systems	2011
Experiments in Robotic Boat Localization	The boat uses sensor-based localization and a follow-the-carrot controller for navigation. Field tests validated its performance in monitoring water quality and handling localization challenges.	The robotic boat accurately followed GPS waypoints, with localization tested under various sensing conditions across 10 km of field trials. Its GPS-dependent design minimizes water disturbance, improving chlorophyll and temperature sampling efficiency	The authors are refining the boat model and implementing a fault detection and recovery framework. Future plans include deploying the boat in real-world measurement campaigns.	IEEE/RSJ International Conference on Intelligent Robots and Systems	2007

Rescue Robot under Disaster Situation: Position Acquisition with Omni-directional Sensor	The paper presents a GPS-free positioning method using angles from an omni-directional sensor with beacons, error minimization, and triangulation. This approach reduces estimation errors and is well-suited for indoor environments.	Simulations confirmed the feasibility of using angle-based positioning for the rescue robot. The proposed network system enables reliable communication and accurate tag localization in collapsed areas.	The system is being implemented on hardware using a PC/104+ board for communication tags. Future work focuses on disaster testing to improve evacuation efficiency using tag-based information.	Proceedings 2003 IEEE/RSJ International Conference on Intelligent Robots and Systems	2003
Efficient Algorithm for Automated Rescue Boats	The paper evaluates search algorithms Breadth-First, A*, and Dijkstra using grid-based simulations for rescue operations. It compares their effectiveness in minimizing cost and time to identify the best algorithm for AI-based rescue planning.	The system uses AI and mobile robots to optimize rescue operations for minimum cost and time, comparing A* and Dijkstra algorithms. A* is faster but less memory-efficient; the system balances efficiency while keeping rescue resources available for multiple tasks.		Proceedings of the 2012 IEEE 16th International Conference on Computer Supported Cooperative Work in Design	2012
Flooding Disaster Oriented USV & UAV System Development & Demonstration	The paper proposes combining GPS and computer vision methods for UAV autonomous landing with an auto take-off and landing system that tracks the mother ship. It integrates UAV and USV into an Aerial and Surface Combined Robot System for improved operational capabilities.	The study addresses rescue challenges in floods, proposing combined USV-UAV systems to overcome USVs' limited view and UAVs' short flight time. It introduces a "double-harpoon system" to improve UAV landing reliability, with experiments planned to evaluate its effectiveness	ChatGPT said: The system is being implemented on hardware using a PC/104+ board for communication tags. Future work focuses on disaster testing to improve evacuation efficiency using tag-based information.	OCEANS 2016 - Shanghai	2016
Visual Pose Estimation of USV from UAV to Assist Drowning Victims Recovery	The paper compares visual localization methods: color thresholding, histogram-based back-projection with Camshfts, and PCA, with histogram methods performing best. All methods were validated through field trials, though PCA showed limitations.	Trials confirmed the localization system's effectiveness for EMILY, with histogram methods outperforming hue thresholding and adapting well. UAV video enabled autonomous navigation to refugees, validated through successful tests.	Future work will enhance object detection using feature tracking and automate parameter tuning for histogram methods. The system aims for full position and orientation integration to autonomously navigate UAVs to victims, speeding responder contact.	IEEE International Symposium on Safety, Security, and Rescue Robotics	2016
On the Design of a Robotic System Composed of an Unmanned Surface Vehicle and a Piggybacked VTO	The paper presents ASV and multi-rotor UAV systems for environmental monitoring, using navigation and coordination algorithms in autonomous and tele-operated modes. Field trials in riverine environments validated the ROS-based hardware and software architecture, with trial data publicly available as ROS log files.	The RIVERWATCH experiment demonstrated effective integration of ASVs and UAVs for robust riverine monitoring, validated through field trials. Datasets are publicly available, with future work focusing on full autonomy, energy harvesting, and improved landing algorithms.	Future work on RIVERWATCH will focus on full autonomy, energy harvesting, UAV charging, and testing landing and smart catch mechanisms for reliability. Characterizing the dual propulsion system and developing new prototypes based on these findings are also planned.	IFIP Advances in Information and Communication Technology Doctoral Conference on Computing, Electrical and Industrial Systems	2014

Cooperative Use of Unmanned Sea Surface and Micro Aerial Vehicles at Hurricane Wilma	The paper explores USV and MAV use in emergency response, emphasizing coordinated operations and real-time spatial reasoning in complex environments. It introduces concepts for USV damage inspection and cooperation with MAVs while addressing challenges near littoral structures.	The first USV-MAV disaster recovery deployment during Hurricane Wilma proved their effectiveness for emergency response and improved situational awareness. The study advanced coordinated robotic operations, highlighted challenges near littoral structures, and suggested new USV-MAV applications.	Future work will address challenges of USVs near littoral structures and improve USV-MAV cooperation. Insights from Hurricane Wilma are expected to inform defense, homeland safety, and new USV-MAV applications.	Wiley Online Library	2008
Flood Disaster Mitigation: A Real-World Challenge Problem for Multi-agent Unmanned Surface Vehicles	The paper presents ASV and multi-rotor UAV systems for environmental monitoring, using navigation and coordination algorithms in autonomous and tele-operated modes. Field trials in riverine environments validated the ROS-based hardware and software architecture, with trial data publicly available as ROS log files.	Feasibility analysis will identify system biases and weights for accurate modeling to improve airboat team performance using qualitative and quantitative metrics. Developing a modular open-source architecture is essential for collaboration and advancing multi-agent systems research in USVs.	Future work involves creating an open-source realistic simulator and prototype vehicle model for testing multi-robot algorithms. The study emphasizes defining challenge problems and evaluation metrics focused on coordination and task allocation in multi-agent systems.	Lecture Notes in Computer Science ((LNAI, volume 7068)) International Conference on Autonomous Agents and Multiagent Systems	2012
Rescue Boat Path Planning in Flooded Urban Environments	The paper uses UAV aerial imaging and image segmentation (grayscale, Canny edges) to map flooded urban areas for rescue operations. It applies path planning algorithms (A*, Genetic Algorithm, PRM) with a cost function balancing distance and obstacle avoidance for efficient rescue boat routing.	ChatGPT said: The algorithms successfully created ground maps from aerial flood images, improving rescue operations with near-optimal boat paths. Path planning results showed trade-offs: A* varied by connections, GA's longer paths with more waypoints, and PRM's shorter paths requiring more computation time	Future work will test algorithms in real-world UAV, rescue boat, and USV cooperative systems for flood rescue. The research aims to improve map building and path planning using additional surface data from USVs with acoustic sensors.	IEEE International Symposium on Measurement and Control in Robotics	2019

2.3 Research Gap

Despite dramatic advances in USV-based flood rescue, there remains some gap to be filled:

- **Real World Validation:** Simulation or small scale demos are dominant, with limited long duration field tests in realistic, debris ridden flood environments.
- **Autonomous Decision Making Under Uncertainty:** Existing planners make assumptions on static or slowly moving obstacles, whereas real flood environments involve rapidly changing debris and currents (Murphy et al., 2008).

- **Sensor Fusion for Localization:** From the literatures it is apparent that , sonar, LiDAR, and visual odometry remains an unexplored areas in terms of flood rescue autonomous boats.
- **Human Robot Interaction:** Reliable connectivity integration in remote operation seems quite unexplored.

2.4 Detailed Review of Methodology

The following points demonstrate the overall design approach, implementation and evaluation appraoch followed in the project:

1. System Design and Integration

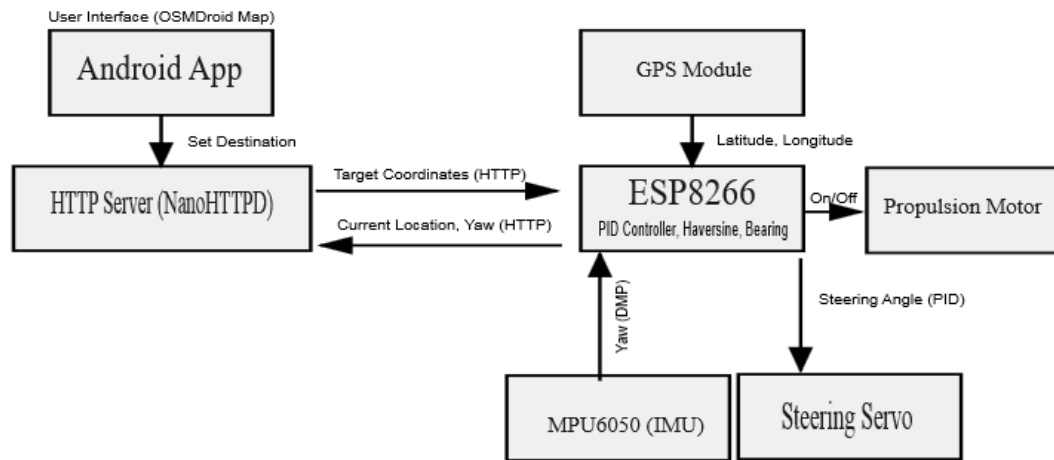


Figure 2.1 Methodology

- **Platform Selection:** A catamaran-style hull was chosen for stability in turbulent, debris filled waters.
- **Sensor Suite:** Integrate GPS, inertial measurement unit (IMU), and forward looking sonar for robust localization and obstacle detection.
- **Communications:** Communication via HTTP sockets over an Android hotspot and Wi-Fi (ESP8266 microcontroller) enables real time telemetry and remote operator override, drawing on principles from multi-agent coordination frameworks.

2.the Destination:. System Architecture

The autonomous boat project integrates two substantial components:

- **ESP8266 Microcontroller (Boat Side):** Manages the boat's navigation by reading data from sensors, driving motors, and talking to an Android application over WiFi.
- **Android Application (User Interface):** Provides a map user interface for a user to enter a target destination, sends these coordinates to the boat, and displays the boat's real-time location and orientation. The system employs an HTTP-based communication protocol in which the Android application plays a server role and the ESP8266 is a client, communicating to enable autonomous navigation.

3. Mandatory Components and Initialization

Boat Side (ESP8266)

- **WiFi Module:** Connects to the Android server using pre-determined credentials (ssid and password).
- **GPS Module (Tiny GPS++):** Reads NMEA data from a GPS receiver through Software Serial (pins GPIO0 and GPIO14) to obtain the current boat latitude and longitude.
- **IMU (MPU6050):** A Digital Motion Processor (DMP) enabled Inertial Measurement Unit provides yaw (heading) data for orienting control, configured over I2C with an interrupt on GPIO13.
- **Servo Motor:** Steers, attached to GPIO15 (D8), 90° center, 30° minimum (left), and 150° maximum (right).
- **Propulsion Motor:** Propels the boat forward, driven over GPIO12 (D6) with simple on/off logic.
- **PID Controller:** Stresses steering to bring the heading of the boat towards aligning with the target direction.

Initialization Process:

- Connecting to WiFi.
- Makes MPU6050 initializes , calibrates accelerometer and gyro, and enable the DMP for quaternion-based orientation data.
- Initializes the servo to the middle position and configures the motor pin.
- Commences communication with the Android server by sending a connect request.
- Android Side
- HTTP Server (Nano HTTPD): Runs on port 8080, accepts requests from the ESP8266, and returns target coordinates.
- Map Interface (OSM Droid): Displays an interactive map (using MAPNIK tiles) where users touch to set destination and view the location of the boat.
- Initialization Process:
- launches the HTTP server when the "Start Server" button is clicked.

- Displays a map centered on a default point (for example, 37.77, -122.41) with level of zoom 10, supporting multi-touch controls.

4. Communication Flow

a) Initialization of the connection:

- The ESP8266 connects to WiFi and sends a connect GET request to the Android server, and the server responds with "Connected to Android Server" and updates the UI to show "USV Connected!".
- Setting
- The user clicks on the OSMDroid map to define a target location.
- The application stores the touched coordinates (latitude and longitude), marks the destination on the map, and transmits them to the ESP8266 via an update lat =.& lon = . GET request.
- Navigation and Data Exchange:
- The ESP8266 gets target coordinates from the server via /update.
- It posts its current location (latitude, longitude) and yaw to the server every 5 seconds via current location lat=. & lon =. & yaw=.
- The app updates the map with the boat position and orientation (yaw rotated marker).
- Arrival at Destination
- Upon arriving at 2 meters to the destination, the ESP8266 stops the motor, aligns the steering and sends a /reached request to the app, updating the UI accordingly.

5. Algorithms Used

5.1 Distance Calculation (Haversine Formula)

- **Function:** Calculates the great-circle distance from the GPS position of the boat to the target position.
- **Implementation:** The Haversine formula is employed by the calculateDistance function, the most appropriate for spherical distances on Earth.
- **Formula:**
- $$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right)$$
$$c = 2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$$
$$d = R \cdot c$$

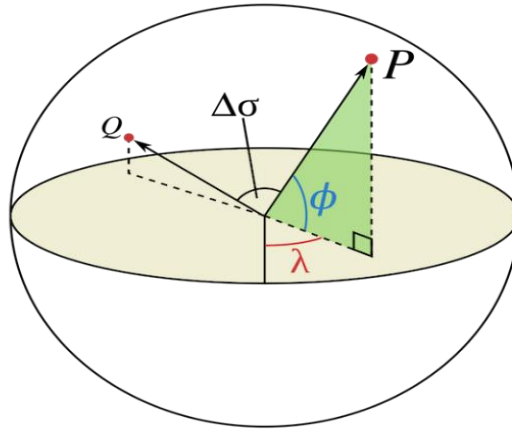


Figure 2.2: Visualization of Haversine formula.[22]

- i. ϕ_1, ϕ_2 : Latitudes of current and target positions (radians).
- ii. $\Delta\phi$: Latitude difference.
- iii. $\Delta\lambda$: Longitude difference.
- iv. R : Earth radius (6371 kilo meters).
- v. d : distance between target coordinate and current location.
- **Output:** Distance in meters.

5.2 Bearing Calculation (Haversine Bearing Formula)

- **Purpose:** To determine the direction (bearing) from the current position to the target, relative to true north.

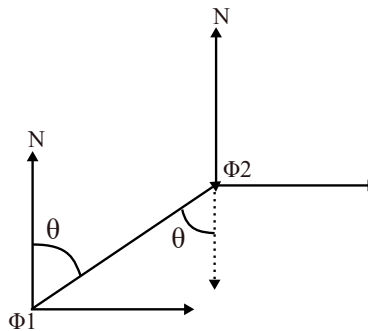


Figure 2.3: Calculating Bearing [21]

- **Implementation:** The calculate Bearing() function uses spherical trigonometry.
- **Formula:** $y = \sin(\Delta\lambda) \cdot \cos(\phi_2)$,
 $x = \cos(\phi_1) \cdot \sin(\phi_2) - \sin(\phi_1) \cdot \cos(\phi_2) \cdot \cos(\Delta\lambda)$,

$$\theta = \text{atan2}(y, x)$$

- Converted from radians to degrees and normalized to 0° to 360°.
- **Output:** θ , Bearing in degrees.

5.3 PID Controller for Steering:

- **Purpose:** Adjusts the steering servo to minimize the heading error between the current yaw and target bearing.
- **Implementation:** The PID Controller function uses Proportional-Integral-Derivative (PID) control:

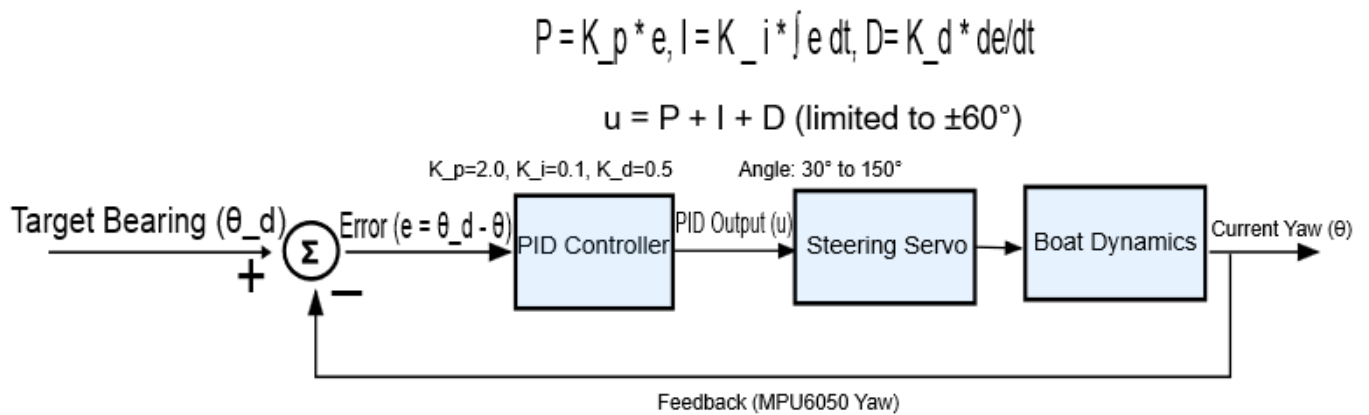


Figure 2.4: PID control loop

- **Proportional (P):** $P = K_p \cdot \text{error}$, reacts to the current heading error.
- **Integral (I):** $I = K_i \cdot \int \text{error} \, dt$, accumulates error over time (with windup protection).
- **Derivative (D):** $D = K_d \cdot \frac{d(\text{error})}{dt}$, responds to the error's rate of change.
- **Parameters:** $K_p = 2.0$, $K_i = 0.1$, $K_d = 0.5$, output limited to $\pm 60^\circ$.
- **Error Calculation:**

Error = target bearing - current yaw, adjusted for wraparound (e.g., if error > 180°, subtracts 360°).

- **Output:** Servo angle adjustment ($\pm 60^\circ$ from center), applied via control Steering.

5.4 Angle Normalization

- **Purpose:** Keeps angles within 0° to 360° and handles wraparound for heading calculations.
- **Implementation:** The normalize Angle function adjusts angles:
 - If angle < 0, add 360°; if $\geq 360^\circ$, subtract 360°.

6. Autonomous Navigation Logic

• Main Loop (loop):

- a. Updates MPU6050 DMP data to get the current yaw.
- b. Reads GPS data for current position.
- c. Gets target coordinates from the server.
- d. Sends current location and yaw to the server every 5 seconds.

• Navigation Steps:

1. Distance Check: If distance to target ≤ 2 meters:
 - a. Sets stop motor (`digitalWrite(MOTOR_PIN, LOW)`).
 - b. Centers servo (90°).
 - c. Sends /reached to the server.
 - d. Disables autonomous mode.
2. Steering and Propulsion: If distance > 2 meters:
 - a. Calculates the target bearing.
 - b. Calculates the servo angle based on the PID controller based on the heading error.
 - c. Limits the servo angle (30° – 150°) and runs it.
 - d. Enables the motor to move forward (`digitalWrite (MOTOR_PIN, HIGH)`).

7. Orientation and Heading Control

- MPU6050 with DMP:
 - Uses quaternions to compute orientation in order to avoid gimbal lock.
 - `tupateDMPYaw` function takes yaw from FIFO packets and converts to degrees, then scales it.

Significance: Proper yaw is critical to accurately orient boat towards target bearing.

8. User Interaction and Monitoring

a. Destination Setting:

1. User taps map, invoking `singleTapConfirmedHelper` to store coordinates and send them to ESP8266.
2. A marker is placed at the target location on the map.

b. Real-Time Display:

- The application `/current_location` updates get a rotated USV marker (through `usv_icon`) at the boat position and yaw value displayed.
- The map is invalidated to update the display.

9. Summary of Methodology

• Initialization:

1. ESP8266 establishes WiFi, initializes sensors and actuators, and configures server communication.
2. Android app starts the HTTP server and map interface.

• Destination Setting:

1. Target is selected by user via the map; coordinates are sent to the ESP8266.

- Typically:

1. Location done via GPS, yaw via MPU6050, and steering done via PID control to the target.

- Monitoring:

1. App tracks boat position and orientation in real-time.

- Algorithms:

1. Haversine formula for distance, bearing calculation for direction, and PID control for steering precision.

This method enables the boat to navigate on its own to any target specified by the user while providing continuous feedback to the Android application, employing robust algorithms for precise control and direction.

2.5 Summary

The USV literature on flood search and rescue highlights five salient areas: firstly, multi agent and heterogeneous teaming (i.e., USV UAV collaboration) for enhanced coverage and flexibility; secondly, advanced path planning algorithms (A*, PRM, GA) that replan in real time for dynamic, debris filled waterways; thirdly, robust sensor fusion of GPS, IMU, LiDAR, sonar and visual odometry for reliable localization even when GPS degrades; fourthly, UAV based visual tracking methods to maintain situational awareness of the vehicle and potential victims; and fifthly, USV–UAV combined platforms that leverage aerial perspectives for far field sensing. Collectively, they demonstrate the promise of USVs to improve safety and efficiency during floods but remain largely restricted to simulation or small-scale tests, underscoring the need for large-scale, in-field validation under real-world flood scenarios.

confined to simulations or small-scale trials, underscoring the need for comprehensive, in-field validation under true flood conditions.

Chapter 3

Design & Implementation

3.1 Introduction

Floods are among the most prevalent and destructive natural disasters, and they are most likely to cause loss of life, property damage, and to communities being halted. This kind of emergencies require search and rescue (SAR) operations to be prompt and effective. Conventional SAR techniques depend highly on human rescuers working in risky and uncertain conditions.

Autonomous Unmanned Surface Vehicles (USVs) are very promising in response to these challenges. USVs are remote-controlled or autonomous water craft that can operate at the without having a human operator aboard. USVs use GPS and communication equipment and can be deployed to track and monitor victims in real-time and deliver emergency if necessary.

The fundamental benefits to the use of USVs on flood SAR missions are enhanced safety, monitoring of deployment area, increased efficiency of operations, and the possibility to reach locations behind debris.

USV usage in SAR is becoming a necessity since global climate change increases the occurrence and the intensity of floods. The project aims to design the capabilities to overcome these challenges by using unmanned USVs in search and rescue missions in floods.

3.2 Problem Statement

Climate change and urbanization coupled with poor drainage infrastructure are the causes of increased frequency and severity of floods. In the event of these disasters, search and rescue (SAR) activities are needed in a quicker manner to find and rescue stranded victims, supply them, and assess damage. But the traditional SAR practices can often put the human responders at risk of dangerous circumstances, like rushing waters, submerged debris, polluted water, and collapsing edifices. Such problems may slow down the time of response and endanger victims and rescuers.

The risk situation is to find a safer, quicker, and more efficient way of carrying out rescue operations in flooded regions, especially in areas that are either not accessible by vehicles or too unsafe to a human being. Although aerial drones have a beneficial impact on surveillance, they are incapable of direct interaction with the water surface and the delivery of heavy loads.

This project will attempt to solve these problems by designing an Unmanned Surface Vehicle (USV) which is specifically meant to help in search and rescue during floods. The USV is to be able to navigate independently or remotely in changing flood conditions, to have real time victim search cameras (video cameras), GPS positioning, avoidance of obstacles, and be able to deliver supplies or tow flotation equipment. The adoption of this system can lead to the increased speed, safety, and effectiveness of response activities regarding floods.

3.3 Design Requirements & Requisites

Designing a USV to search and rescue during floods has to take into account a set of mechanical, electrical, software, and environmental parameters to allow efficient work in real-life conditions of disaster situations. The main design requirements/ requisites are as follows:

1. Buoyancy and Stability

The USV should be capable of floating and being balanced in turbulent and in debris-filled flood water. The design of the hull (e.g. catamaran or flat-bottom) should be stable so that it is more maneuverable and stable.

2. Propulsion and Navigation

- Powerful propulsion should counteract any powerful currents and shallow water.
- Waterproof, energy efficient, electric motors are preferred.
- The car must be capable of autonomous and manual (remote-controlled) navigation.
- Precise localization and path planning is necessary, so GPS and IMU (Inertial Measurement Unit) integration is needed.

3. Obstacle Detection and Avoidance

- Ultrasonic, LiDAR or infrared sensors to sense floating debris, submerged objects and walls.
- Safe navigation should be done with real-time obstacle avoidance algorithms.

4. Search and Rescue Equipment

- Cameras (visual, and /or thermal) to detect victims in water or roofs.
- Loudspeakers or lights for communication or signaling.
- Hooks, ropes, or flotation devices for rescue or payload delivery.

5. Power System

A long battery life to sustain long missions.

Rechargeable battery packs or solar support option: long run time.

6. Durability and Waterproofing

- The whole system is to be water-resistant or waterproof (IP67 or more), the motor, electronics, and sensors.
- Anti-corrosive and anti-dust that is usually present in floodwaters.

7. Communication

- The connection to the control station in real time via wireless protocols (e.g., Wi-Fi, RF,, LoRa etc.).
- Redundant communication options for fail-safe operation.

8. Portability and Deployment

- The USV must be portable and small to carry and be deployed easily in times of emergency.
- There might be need of quick assembly/disassembly in field conditions.

9. User Interface

A simple, intuitive control interface for operators with minimal training.

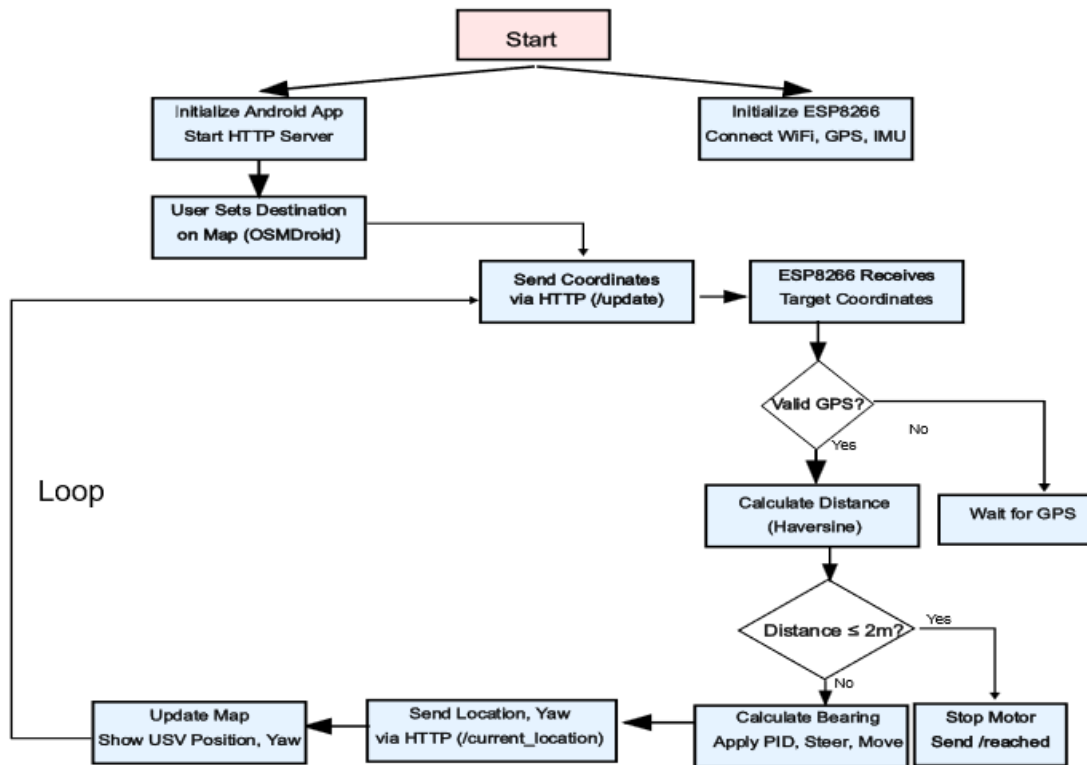


Figure 3.1: System Flowchart.

Live data feed (video, GPS location, battery status) to the ground station or mobile app. Designing a USV to search and rescue during floods has to take into account a set of These conditions guarantee that the USV will be able to adequately serve search and rescue operations as it is able to perform successfully in unpredictable and dangerous flood conditions.

These requirements ensure the USV can effectively support search and rescue missions by operating reliably in unpredictable and hazardous flood environments.

3.4 Design Criteria

Design specifications detail the measurable standards and performance criteria by which the USV will be considered operational, safe, and effective for flood search and rescue operations.

1. Performance Operation

- Speed: At least 2–5 km/h in calm water; must be able to navigate itself in moderate currents.

- Range: Must be able to travel at least 2–5 km per mission.
- Payload Capacity: Must have at least 3–5 kg capacity for equipment, flotation devices, or rescue equipment.

2. Maneuverability

Small turning radius to navigate across obstructed or debris-strewn areas.

Reverse or fast braking functionality during emergencies.

3. Autonomy and Control

- Manual remote operation or autonomous mode assistance.
- GPS real-time tracking with error allowance within ± 2 meters.
- Fail-safe home return functionality on signal loss.

4. Sensing and Detection

- Low-resolution vision camera of at least 720p.
- Optional thermal vision for person detection in poor visibility.
- Obstacle sensors with a minimum of 1–2 meters.

5. Power and Durability

- Minimum 60–90 minutes' battery life for continuous operation.
- Quick-charging facility or exchangeable battery packs.

6. Safety and Reliability

- IP67 or higher waterproof rating.
- Resistance against capsizing in moderately rough seas.
- Duplicated communications systems to prevent loss of control.

7. Ease of Deployment

- Weight below 10 kg for manual lift by 1–2 persons.
- Rapid boot and calibration process (under 5 minutes).

8. Cost and Maintainability

- Use of low-cost, readily available components.
- Modular design so components can be readily repaired or replaced.

These specifications ensure that the design of the USV is feasible, effective, and deployable under real flood rescue situations.

3.5 Design Steps

1. Problem Analysis & Requirement Gathering

Flood rescue operation problem was decomposed. System requirements and constraints were delineated based on real rescue scenarios. System objectives such as victim detection, supply delivery, and safe navigation through dangerous conditions were determined.

2. Conceptual Design

Various hull types were considered, and the suitable design (e.g., catamaran or flat-bottomed) was selected based on stability within the floodwater. Initial system configurations, including the location of motors and sensor locations, were sketched. Propulsion approaches were evaluated based on water depth, flow, and debris conditions.

3. System Architecture Design

The system was divided into main subsystems: electric system, sensing unit, control unit, communication module, and mechanical structure. A modular design approach was used to simplify integration and development in the future.

4. Component Selection

Components that were compliant were selected based on performance, cost, and availability. They included brushless DC motors, electronic speed controllers (ESCs), GPS modules, ultrasonic sensors, cameras, microcontrollers (e.g., Arduino or Raspberry Pi), and waterproof batteries.

5. Mechanical Design and Fabrication

Hull and frame were modeled on CAD software and constructed from lightweight, water-resistant materials such as marine-grade plywood or HDPE. Motor, sensor, and electronic mounting points were incorporated in the design. The structure was sealed and reinforced for waterproofing and stability.

6. Electrical Integration

The power system was connected, and electronic components were integrated. Wiring was organized and housed in waterproof enclosures. Redundancy aspects and wet environment safety measures were implemented to protect the system from water.

7. Software Development

Control algorithms were programmed for manual (remote-controlled) and autonomous modes of operation. Sensor data processing, GPS navigation, and obstacle avoidance features were included. A simple user interface was optionally implemented to display real-time telemetry.

8. Testing and Calibration

Initial testing was performed on land to validate wiring, motor function, and software reaction. Water tests followed in controlled environments to validate navigation, stability, and obstacle detection. Sensors and control elements were tuned for accuracy.

9. Performance Testing

System performance was recorded in terms of speed, endurance, maneuverability, payload capacity, and communication range. Vulnerabilities discovered during testing were addressed through design modification and optimization.

10. Final Optimization and Deployment

It was refined according to test performance in regard to efficiency, strength, and ease of use. The finished system was field-packaged, and user manuals or training schedules were established to optimize its application in real flood rescue missions.

3.6 Design Assessment & Performance Indication

To examine the performance of the designed Unmanned Surface Vehicle (USV) in search and rescue operations for floods, a set of performance indices was identified and compared through a thorough analysis. These indices were chosen taking into account the mission requirements, environmental conditions, and favorable operating conditions.

1. Speed and Maneuverability

The maximum speed of the USV was tested in cases of still and running water. Turning performance and response to control inputs were found to navigate in narrow or blockaded routes. The performance was accepted when the vehicle was able to maintain stable navigation at speeds between 0.5-1 km/h.

2. Endurance and Range

The range and battery life of the USV was checked under normal loads. These performance levels were set as minimum runtime 60 minutes and range 1-2 km. Endurance was checked by running continuously and was validated from observations of battery discharge rates.

3. Payload Capacity

The system was also subjected to testing for the ability to carry rescue equipment, life jackets, or first aid kits. Payload capacity was determined by incrementally adding load until stability or propulsion efficiency was compromised. The minimum payload was 0.5-1 kg.

4. Stability and Buoyancy

Evenness of loading was maintained at all times, and stability in uneven loading and turbulent water was tested. The center of gravity and buoyancy distribution of the vehicle were simulated to avoid capsizing or tipping under operating conditions. Dynamic weight shifts and partial submersion conditions were tested.

5. Navigation and Control Accuracy

Precision in GPS-guided navigation and autonomous path-following software was verified. Planned route deviation was monitored, and error tolerance was established at ± 5 meters. Responsiveness of remote control manual control was also established.

6. Communication Reliability

The functionality of wireless communication (i.e., RF or Wi-Fi) between the control station and the USV was assessed in terms of range, data latency, and loss of signal. A 100–300 meter range for control was considered acceptable.

7. Ease of Deployment

Portability, simplicity of use, and setup time were also tested. Setup time below 5 minutes and total weight of the system below 5 kg were considered optimal for usability in the field.

3.7 Summary

To neutralize the increasing severity and frequency of flood cases, a low-risk yet efficient alternative to search and rescue missions has been proposed in the form of an Unmanned Surface Vehicle (USV). The system was proposed to traverse unsafe, debris-ridden, and water-saturated areas where normal rescue missions are imminent or unsafe. In the design stage, critical specifications such as stability, maneuverability, payload carrying capacity, autonomous navigation capability, obstacle avoidance, and real-time communication were considered with accuracy. The USV is designed to be as simple, efficient and cost effective as possible. The principle and operational algorithm is also kept efficient to ensure low power operation of MCU.

Chapter 4

Project Management & Finance

4.1 Introduction

Proper planning and proper financial strategy are important in the successful implementation for any engineering project. Careful-planning, well thought out scheduling, budgeting, and resource collection played a important role in the successful completion of the Unmanned Surface Vehicle (USV) project. It was a project involving the coordination of various phases of component selection, circuit design, mechanical assembly, programming and testing of the same within a limited time and budget.

The project demanded careful selection of complexity in build materials, selection of financial plan due to the limited budget constraint. At the end of the day giving utmost importance to the timely execution of plans and dedicated work made the project successful

4.2 Management of Project

The project saw success by effective planning and smooth implementation with working in patience without indulging in a hurry. The development can be described by points like component choice, hardware development, coding, and testing. The team members properly followed the lead of the team lead that avoided any conflict of interest among team members. Components were carefully selected based on functionality, affordability , effectiveness and cost. Continuous communication and meeting among team members ensured proper guidance in every problem steps.

4.3 Time Frame / Gantt Chart

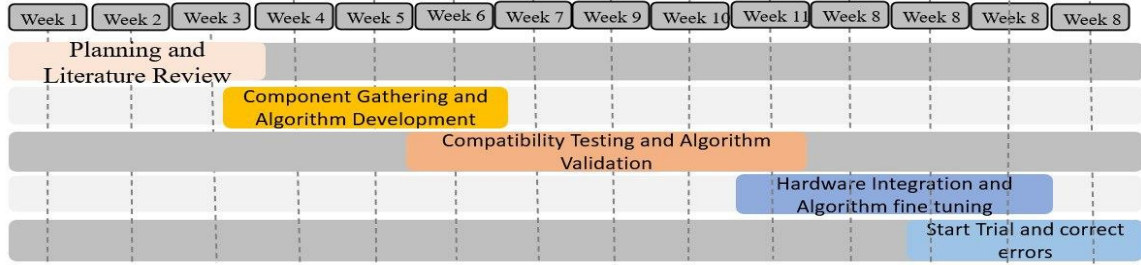
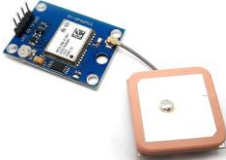
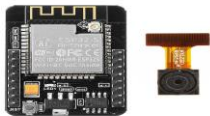
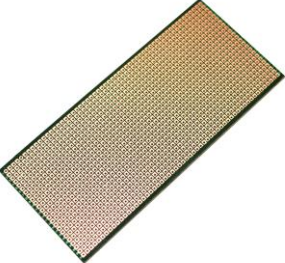


Figure 4.1 Gantt chart for time frame

4.4 Equipment Selection

Table 4.1 Equipment Selection

1. ESP32 microcontroller 	2. GPS Module NEO6MV2 
3. Motor driver tb6612fng 	4. DC motor 
5. SG 90 servo 	6. ESP 32 CAM 

7. Gyroscope GY 521 MPU 6050 	8. MOSFET 
9. Cork sheet 	10. Veroboard 
11. Battery 	12. Propeller 

4.5

Overall budget

Table 4.2 Overall budget

Components	Description	Quantity	Price (Tk)
ESP32 microcontroller	Wi-Fi-enabled microcontroller	01	600
GPS Module NEO6MV2	GPS location tracking module	01	800
Motor driver tb6612fng	Dual DC motor controller	01	250
Battery	Power supply source	3x1 cell	1550
DC motor	Rotational actuator	01	100
SG 90 servo	Small positional servo motor	01	230
Gyroscope GY 521 MPU 6050	Gyroscope and accelerometer sensor	01	208
Cork sheet	Lightweight floatation base	01	100
MOSFET	Electronic switching component	01	150
propeller	Generates thrust in water/air	01	280
Veroboard	Circuit prototyping board	01	150
Total Cost			4418

4.6 Summary

This chapter described the overall planning financial management of the successful completion of the Unmanned Surface Vehicle (USV) project. The 8 week schedule was strictly followed to successfully complete the tasks that were a must for the completion of the project. Cheap but efficient components were selected to balance the trade-off between price and performance. Strong communication between teammates ensured efficient distribution of workloads crucial to reach all the technical goals.

Chapter 5

Addressing Complex Engineering Problem & Activities

5.1 Introduction

Engineering projects are generally multidisciplinary and disorganized in nature. This project was no exception; it was done by solving various engineering problems. Problems that require specialized knowledge and expertise to solve. Our project Unmanned Surface Vehicle (USV) included mechanical, electrical, mathematical, and computer science models and their integration to work properly. Other skills like critical thinking, a creativity, expertise with tools, guessing future outcomes etc. were all required to solve the required engineering problems and to complete the project. The required engineering problems and their solution methods are mentioned and explained in the following tables. Each of the engineering problems criterias are based on the BAETE OBE manual.

5.2 Addressing Complex Engineering Problem

Table 5.2.1: Complex Engineering Problems with description.

Class	Topic	Description
P1	Depth of Knowledge Required	Implementation required advanced knowledge in OOP in C++ and Kotlin programming languages, Global Positioning System (GPS), IMU sensors, sensor fusion and Mechanical propelling methods.
P2	Range of Conflicting Requirements	Reliable and accurate operation along being cost effectiveness and environmental suitability.
P3	Depth of Analysis Required	Required solid problem solving skills by integrating totally different domains like programming with Navigation, mechanical propulsion and implementation of working circuitry.
P4	Familiarity of issues	Tackled previous unfamiliar topics like Global Position System, Calculation of bearing angle using Haversine Formulae.
P5	Extent of applicable codes	Followed ethical and legal factors ensuring compliances.
P6	Stakeholder Involvement & Conflicts	Balanced contradictory needs of high performance in adverse conditions along with limited cost
P7	Interdependence of System components	Balanced seamless communication between Android control Interface with MCU of USV. Ensured Robust and error compensated navigation using PID algorithm utilizing sensor feedback.

5.3 Addressing Complex Engineering Activities

Table 5.3.1: Complex Engineering Activities.

Clause	Topic	Description	Remarks
A1	Range of Resources	Used various sensors like IMU, GPS along with MCU-ESP8266 etc. Implemented Code using Android Studio and Arduino IDE.	Proper integration improves navigation and rescue efficiency.
A2	Level of Interaction	Ensured seamless communication between Android Control Interface and MCU of USV. Extracted optimal-low power performance of components across the project.	Advanced efficient communication algorithm requires tedious and time consuming effort. Low cost implantation requires extensive research for suitable parts.
A3	Innovation	Designed super-efficient navigation algorithm using C++ along with efficient communication algorithm that is run without lag on dirt cheap ESP8266 MCU. Which in turn, enabled implementation of super low cost Autonomous USV.	Applied mid-level language (C++) with efficient algorithm to achieve super cheap and reliable autonomous USV solution.
A4	Societal & Environmental Impact	Aims to use low-power and easy to control USV for rescue missions with ease, saving time and reducing reliance on environmentally harmful engine boats. Also uses biodegradable chassis (cork-sheet).	Addresses environmental sustainability with low cost solutions and environmentally suitable materials.
A5	Familiarity	Implemented and made fusion of existing techniques to innovatively generate reliable, cost-effective USV.	Usage of conventional techniques creates for easy upgradation in the future

Chapter 6

Result & Analysis

6.1 Introduction

For evaluating the capabilities of the USV, we performed several field tests (or should say water test). These tests were done to confirm the capabilities of the USV and whether it reaches the minimum performance threshold.

We evaluated key parameters such as speed, turn radius, connection range with the controller, payload capacity, battery life etc. Each parameter was tested more than once and the average values were taken. We recorded these data and then evaluated to see if it meets satisfactory result.

We verified our results and performance parameters with similar works/devices, found in the literature review. We deemed the comparison satisfactory and hence determined the capabilities of the vehicle.

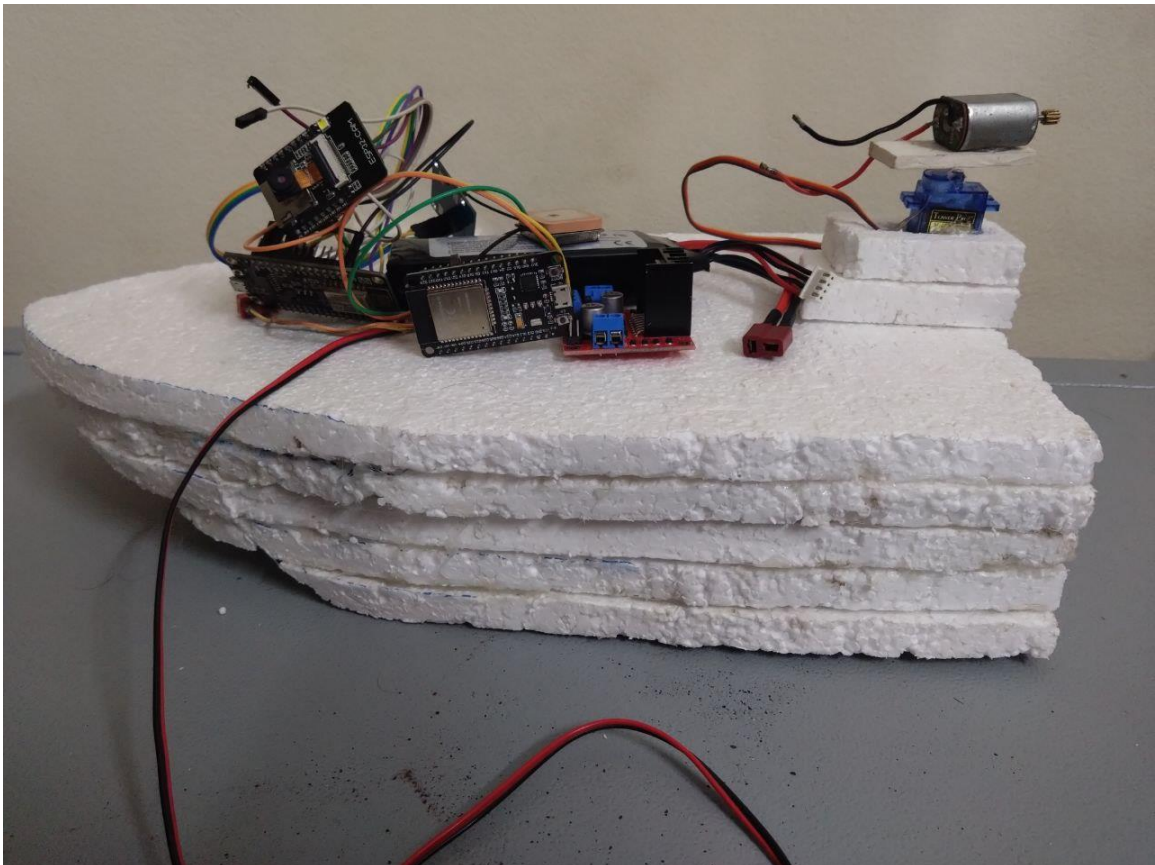


Figure 6.1 Constructed model

6.2 Result & Analysis

This is the table compiling the tested performance parameters for the Unmanned Surface Vehicle (USV). The parameters were determined after multiples tests and only the average was then taken to ensure any environment effects are eliminated. These performance metrics aims to present the capabilities and usefulness of the USV in complex search and rescue environments.

Table 6.2.1: Result table with performance specifications.

Parameter	Measured Value / Description
Average Speed (Calm Water)	1 km/h
Average Speed (Flowing Wate	5 km/h
Turning Radius	2 m
Operation Duration	40 minutes
Distance Covered	0.5 kilometers
Payload Capacity	Up to 0.5 kg (no stability/efficiency loss)
Sensor Performance	Affected by water turbulence
GPS Accuracy	± 1.7 meters
Communication Range	260 meters (stable with minimal packet loss)
Stability	Maintained under varying loads

6.3 Limitation

As it is a project for demonstration purpose, so it has various limitations. These limitations include low structural integrity (for using cork sheet), high center of mass, wide body (increases stability but decreases speed), control system is not robust, low battery capacity, signal requires clear Line-of-Sight with the satellites above (for gps navigation), Thus, the sensor ability of the vessel restricts the operator of the vessel. It is also a surface vessel and hence higher risk of getting stuck as it navigates through narrow conditions.

6.4 Summary

The Unmanned Surface Vehicle (USV) designed for flood search and rescue operations successfully met the primary objectives of stability, maneuverability, endurance, and payload capacity. Testing demonstrated that the vehicle could operate reliably in simulated flood conditions, achieving speeds suitable for timely rescue efforts and navigating through constrained and debris-laden waters. The autonomous navigation system, supported by GPS and obstacle detection sensors, enabled effective path following and real-time obstacle avoidance with acceptable accuracy and responsiveness. Communication systems provided robust control and telemetry over practical distances. We validated the waterproofing of the vehicle and ensured sufficient durability for the demonstrations. Some minor improvements such as sensor upgrading and upgraded controller and GPS module can be adopted. Also AI can be integrated if better controllers are adopted.

Chapter 7

Socio-economic Impact on Sustainability

7.1 Introduction

Engineering project performance now also require impact to society, economy and the environment. As the significance of sustainable development increases, social responsibility is becoming vital for acceptability. In development of new engineering solutions, innovations that introduce social responsibility and sustainable development are valued in the current world. The Unmanned Surface Vehicle (USV) project is such a flood search and rescue Device. Its design considers the socio-economic and environmental impact of floods on disaster rescue services. Floods are particularly devastating particularly to the economically poor population. The floods destroy infrastructure, interfere with economic activities, and crush emergency services. This USV project aims search and rescue missions in a cheaper, safer and more reliable way. This project also targets a positive social effect by minimizing human risk, enhanced worker safety and lowering operational costs, faster reaction to rescue needs.

This project will also be sustainable due to the application of low-power electronics that reduce the power consumption and improves performance in areas affected by floods.

7.2 Impact on Societal, Health, Safety, Legal and Cultural Issue

Societal Impact:

The Unmanned Surface Vehicle (USV) project meets the short-term requirements in the flooded regions especially where fast and safe responses are immediately required to save human lives. It provides efficient and affordable responses to emergencies. It can navigate in problematic locations where human navigation have safety concerns. In conditions of low visibility, the USV can be used instead of traditional means of rescue effort. It enables the human rescuers to make decisions and provide care working on areas that are important like scanning and identification of victims. The USV can also be used to other types of service, such as environmental monitoring, which is an added advantage. In addition, it introduces computerized disaster management infrastructure and support the community to build technologically oriented future by reducing the number of people working manually. It can enhance the development of smart cities and digital disaster systems and assist communities in shifting to resilient and tech-oriented future by reducing manual intervention.

Health and Safety:

In addition to minimizing the risk of life-threatening risks, including drowning or electrocution, non-invasive search functions of the USV do not lead to unnecessary disruption of sensitive rescue assets, such as partially collapsed structures or floating debris. It can also possess environmental sensors, including pollutants, gas leakages or bacteria sensors. This helps the health authorities identify the secondary health dangers on the spot. Ability to provide medical kits, flotation devices or emergency food packets can help to survive easily until a rescue mission is carried out. On the psychological front, the realization that such a technology was used can enable the victim to calm down and assure her that there are people working on the rescue operation.

Legal Considerations:

The legal considerations, other than compliance and privacy issues, should also include the protection of custom USV design intellectual property, in particular, when the University, NGO, or government agencies are involved. Liability must also be created, e.g. pre-identified responsible parties in case of mission failure, or in case of damage caused by the USV. To incorporate USVs in emergency management systems through SOPs, ethics, autonomous action authorization procedures and autopilot operation approved frameworks they should be allowed by law. A law also should exist to undertake expedient legislation when deploying USV in times of emergency that meander through bureaucracies after the declaration of certain disasters.

Cultural Acceptance:

The long-term adoption will be dependent on the utility that individual assign to it and its relationship with the community. The USV need to be viewed as not just a technological tool, but a life-saving partner, which the community helps to develop. Involve the youth in the locality or volunteers in it's presentation and it operation will create a sense of ownership and pride. In addition, design aspect (e.g. color, signage, even onboard announcement) could also be localized to the local identity and language. This make the USV less alien. In the time of deployment, it is important to note that the locals will be treated respectfully in relation to local practices that could be delicate in locations where visual surveillances is necessary. This helps to prevention of the cultural issue. In other contexts, it may be helpful to present story and reviews of previous users for build trust and for make the automated system seem more suitable of response for a crisis.

Educational and Skill Development Opportunities:

The implementation of this highly developed systems stimulates knowledge and training in the STEM areas. Learners, technicians and local first responders have the opportunity to immerse in robotics, communication technology, and disaster response and acquire hands-on experience. The USV project by collaborating with schools and universities can serve as an accelerator of new concepts and studies to motivate the new generation of innovators that are willing to tackle the challenges of tomorrow.

7.3 Summary

The socio-economic and sustainability effects of the Unmanned Surface Vehicle (USV) intended for flood search and rescue were examined in this chapter. The USV fulfills the need of the societal demands that are crucial because of the possibility of applying a safer and more efficient alternative to traditional rescue methods, especially on remote or submerged sites. It reduces the time to respond and reduces health hazards to human rescuers using real-time location data and sensor data. Besides promoting social gains, the project promotes sustainable development through the use of low-power electronics and affordable components, and it is therefore affordable with low resource communities. Responsible deployment was also brought up along with legal and ethical concerns such as privacy and regulatory compliance. Also, cultural acceptance is a critical element in the achievement of successful incorporation of such technology. On balance, the USV project demonstrates that innovative engineering can contribute to sustainability goals and leave a great impact on the society.

Chapter 8

Conclusion

8.1 Introduction

The project is a real-life example of how smart systems could be implemented during the disastrous events to conduct and provide support with the rescue operations. Although UAVs (Unmanned Aerial Vehicles) have better mobility, and more coverage area on the ground, they are unable to perform rescue or relief missions, as they are restricted by their own weight. It is the ideal situation in which a USV (Unmanned Surface Vehicle) can be more of a useful thing. As an autonomous floating platform, it has no physical weight limit, meaning that relief supplies (food, medicine etc.) are transported at a higher efficiency by these USVs. In addition, in case of urgency, the sufficient USV size can be deployed to perform the extraction operation independently, without the need to wait until a rescue party arrives, to provide the affected person with a better chance of survival. With this USV, the operators can control the USV accurately since the USV can determine its position using the satellite navigation (GPS). In case no latest satellite imagery is available, the vehicle is able to use its cameras to permit a visual navigation. This comes in handy when one intends to search a victim within a particular area.

8.2 Future Research Direction

The present prototype of the USV demonstrates good potential in flood rescue, and a number of directions provide opportunities to conduct research and improve it:

1. **AI-Based Victim Detection**

In the future, it is also possible to make computer vision and AI models available to identify victims through visual and thermal imaging, specifically in the low-visibility environment.

2. **USV-UAV Coordination**

The use of USVs to coordinate with UAVs on search missions can prove useful in expanding the coverage of areas and real time environmental awareness.

3. **Enhanced Sensor Fusion**

LiDAR, sonar and visual odometry and GPS and IMU will also be added to the system to improve localization and the detection of obstacles in dynamic waters with debris.

4. **Swarm Operations**

A coordinated swarm of USVs would be able to accelerate the coverage of rescue and enhance performance in extensive flood areas.

5. **Real-Time Adaptive Path Planning**

The advanced algorithms can make the USV modify its trajectory to any emerging obstacles or conditions (Genetic Algorithms and Reinforcement Learning).

6. **Extended Field Testing**

Long terms trials and testing is needed to ensure reliability of the system under various conditions and environments.

7. **Improved Communication Systems**

Complex and sophisticated algorithms are used, these allows the USV to adjust itself to any environment and overcome problems.

8. **Societal and Legal Considerations**

This USV will have to checked by regulators for licensing for ensuring safety during operation.

These criteria's can be improved upon by future research and development.

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